7. $6y = \left(e^2 + e\right)e^x - \left(e^2 + e + 1\right)e^{2x} + e^{4x}$ 8. $ye^{\sin x} = 2e^{\sin x} + c$

9. $\log y = \frac{x^3}{3y^2} + c$ 10. $\log|1+y| = x + \frac{x^2}{2} + c$

5. Numerical Methods

Exercise 5.1

1. $\log\left(1 + \frac{h}{x}\right)$

2.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0	-1					
		1				
1	0		4			
		5		6		
2	5		10		0	
		15		6		0
3	20		16		0	
		31		6		
4	51		22			
		53				
5	104					

5. $\frac{-2}{(x+1)(x+2)(x+3)}$ 6. 31 7. 445 lakhs 8. 3 and 24

Exercise 5.2

1. 6.8 2. ₹2,900 3. $f(x) = 2x^3 - 7x^2 + 6x + 1$
4. 36.784 (lakhs) 5. 197 6. 15.45
7. 286.96 8. 27.992 9. 108.75 (Thousand tones)
10. 41 persons 11. 476.25 Lakhs 12. 53

Exercise 5.3

1	2	3	4	5	6	7	8	9	10	11	12	13	14
(a)	(c)	(a)	(c)	(d)	(a)	(c)	(c)	(c)	(c)	(a)	(c)	(b)	(b)

Miscellaneous Problems

3. $f(x) = x^2 - 3x + 1$ 4. 14.25, 23.5 5. 128.5
6. 189.79, 286.96 7. 5281, 6504
9. $y = \frac{2}{3}x^4 - 8x^3 + \frac{100}{3}x^2 - 56x + 31$ 10. $y = x^3 - 8x^2 + 19x - 12$

XII Std - Business Maths & Stat EM Answers.indd 272 2/28/2020 1:40:59 PM